

Energy-Efficient Llama3 Acceleration on a CGLA by Offloading Computational Kernels

Takuto Ando, Yu Eto, Ayumu Takeuchi, Yasuhiko Nakashima

Nara Institute of Science and Technology, 8916-5 Takayama-cho, Ikoma, Nara 630-0192, Japan.

Background & Objective

The LLM Energy Challenge

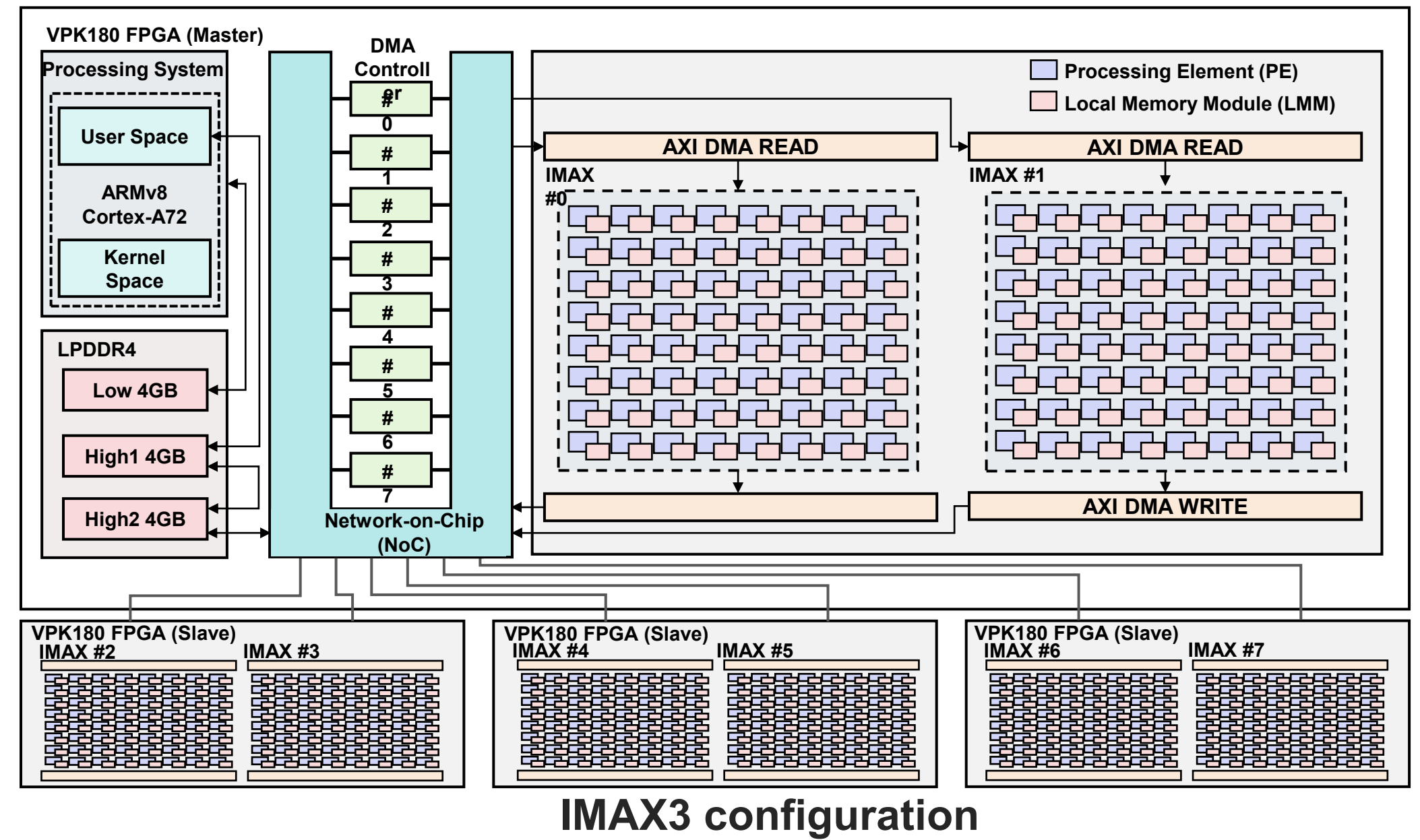
The rapid evolution of Large Language Models (LLMs) results in massive energy consumption, demanding new, power-efficient hardware solutions beyond traditional GPUs.

IMAX3 Accelerator

IMAX3 is a CGRA architected with a linear array of processing elements and on-chip memory[1].

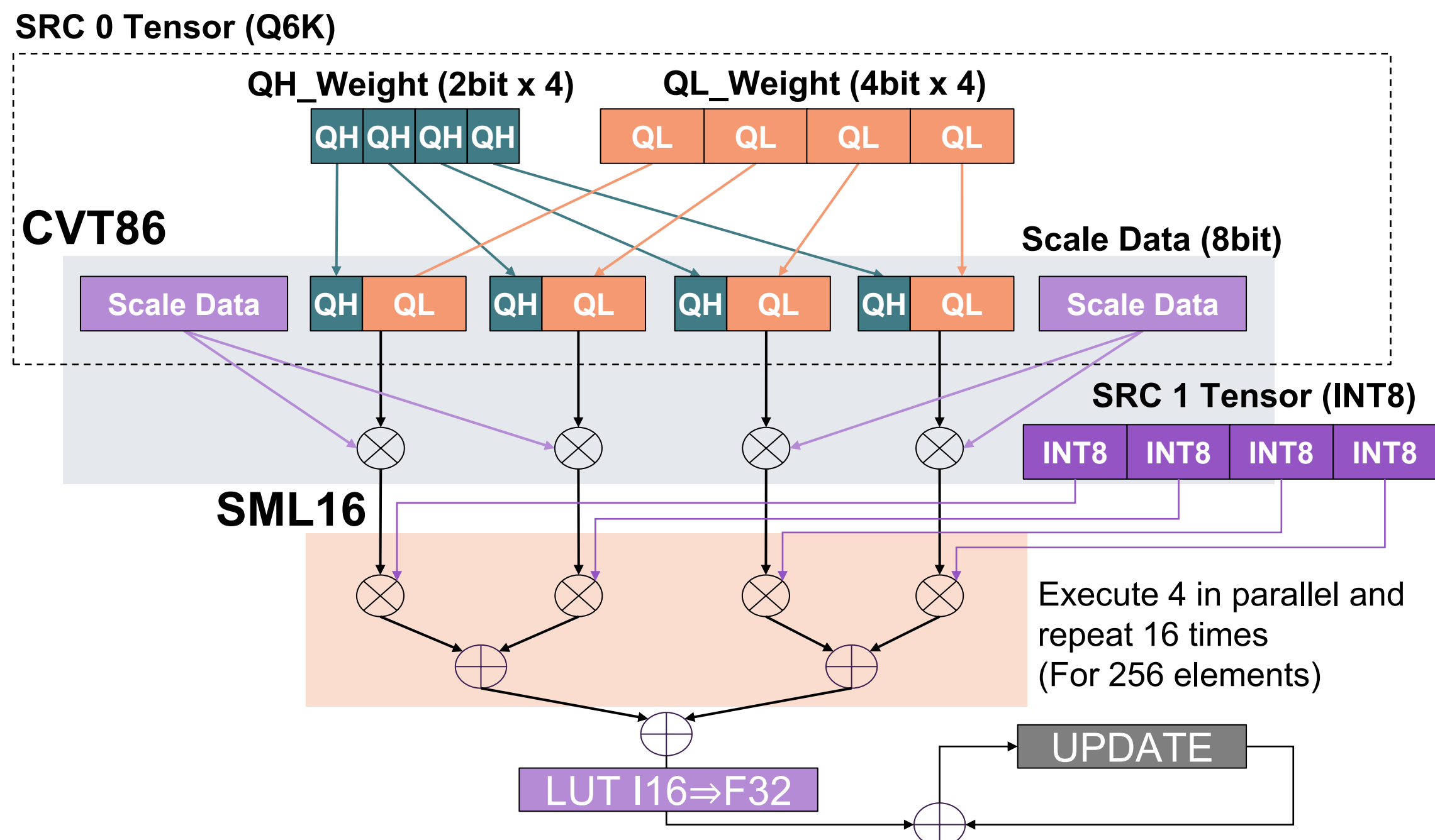
Objective

To accelerate the latest Llama3 model by offloading computationally intensive Q6_K kernels directly onto the IMAX3 hardware to maximize performance.



Methods & Results

Offloading the Q6_K Kernel to IMAX3



The kernel's dataflow is optimized for the IMAX3 architecture and leverages two novel custom instructions for efficient processing:

- CVT86:** This instruction decodes the complex input format, converting the 2-bit/4-bit quantized weights and 8-bit scale data into a unified 16-bit intermediate format ready for computation.
- SML16:** Following the decoding step, this instruction performs a highly parallel dot-product operation between the 16-bit intermediate data and the 8-bit integer input.

This targeted optimization successfully removes the need for host CPU intervention during these calculations. The hardware offload ratio is increased from approximately 90% to nearly 100%

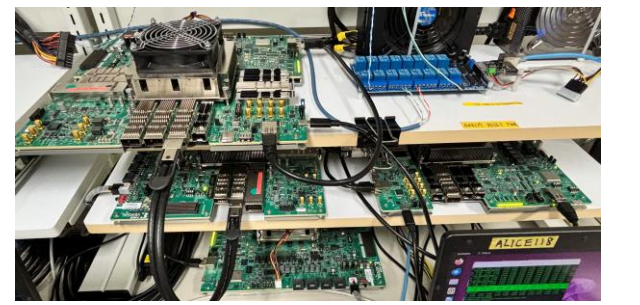
Experimental Setup

Model: Llama3-8B Q3_K_S

Task: Generating 32 tokens

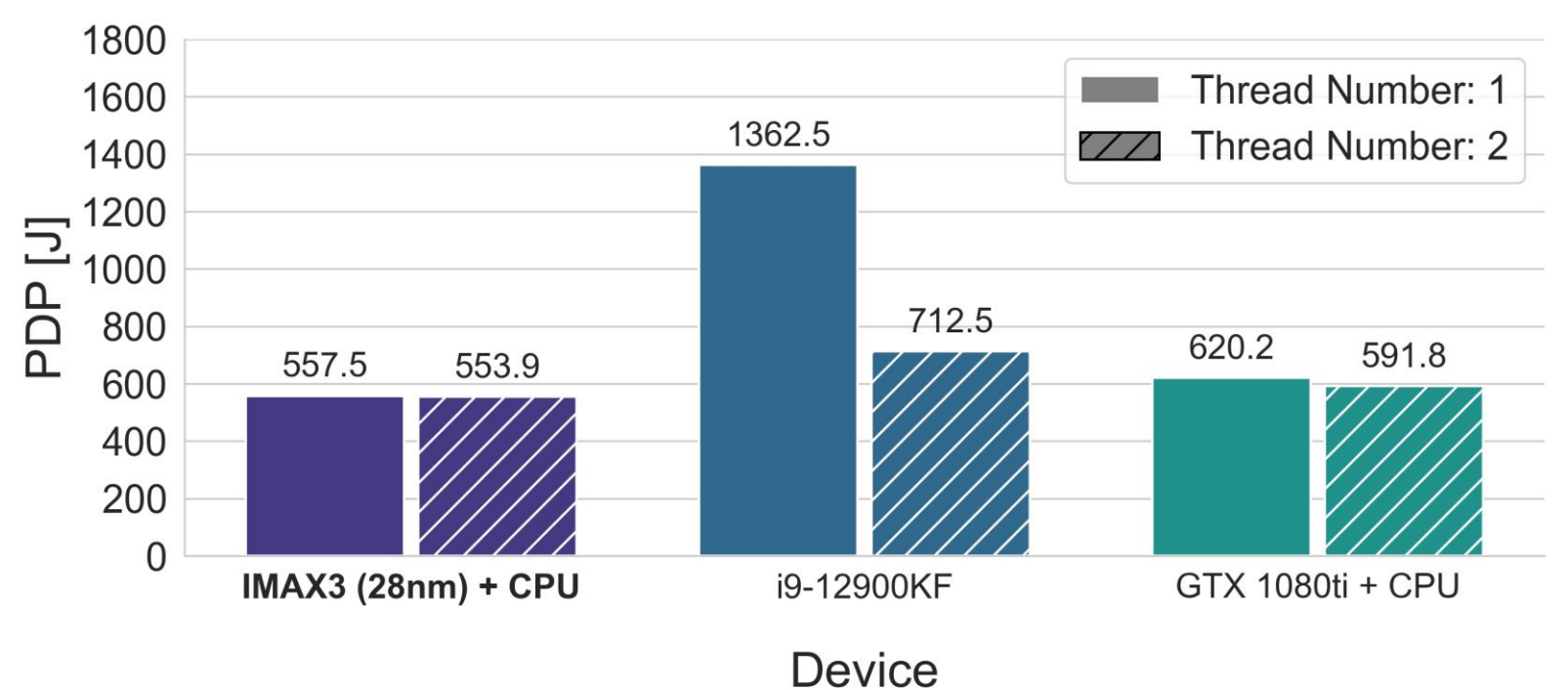
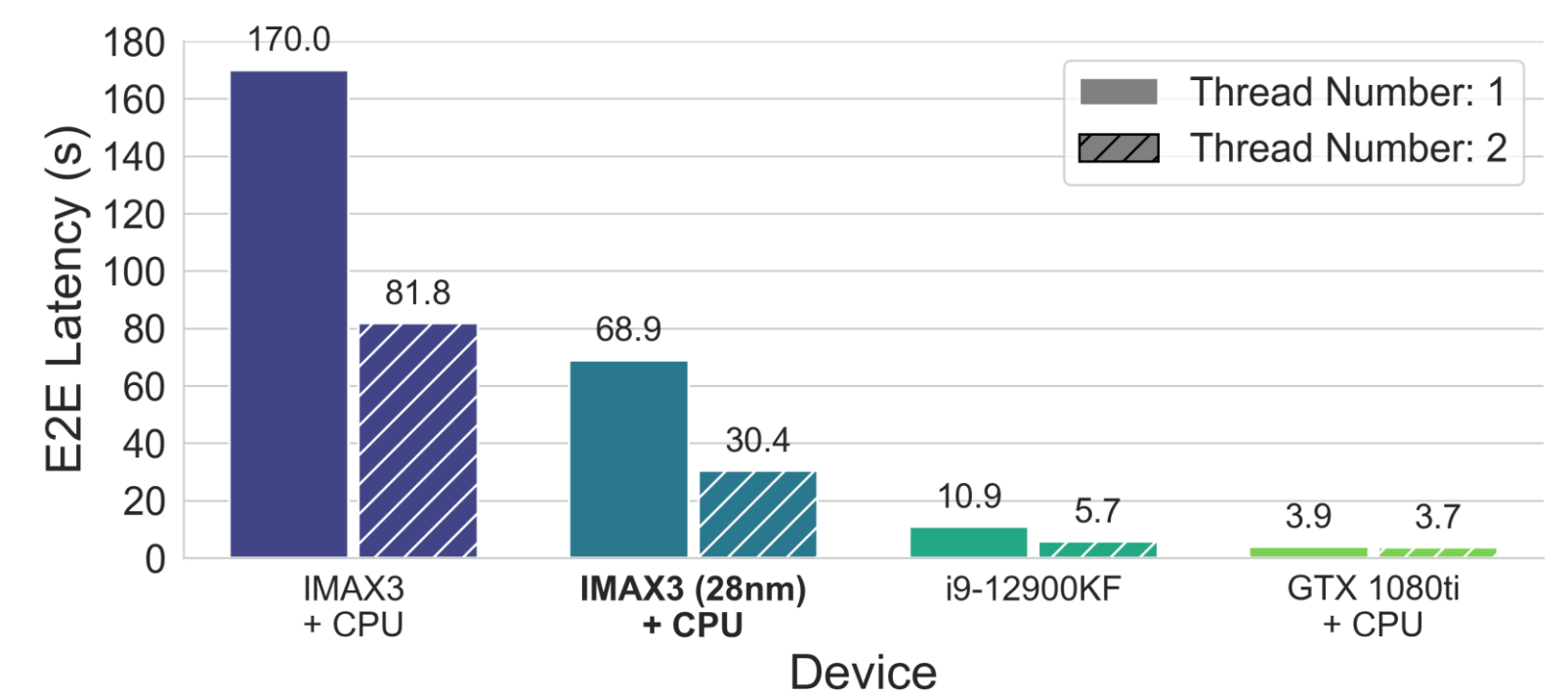
End-to-End Latency: To measure absolute processing speed.

Power-Delay Product (PDP): Calculated as Latency $\times$ Power



Device	Cores	Chip area (mm ²)	Process node (nm)	Frequency (MHz)	Power (W)
IMAX3 (Xilinx VPK180)	64	-	7	140	180
IMAX3 (28nm)	64	14.6	28	840	25.4 or 31.7
Intel i9-12900KF	16	-	7	2400	125
NVIDIA GTX 1080 ti	3584	471	16	1480	250

Evaluation Results



GPU remains the low for E2E latency

IMAX(28nm) is more energy-efficient for PDP

Discussions

Performance Improvement from Q6_K Offloading

Strategy	Offloaded Kernels	Latency (32 Tokens)	PDP [J]
Previous Work	Q3_K only	34.2 s	464
This Work	Q3_K + Q6_K	30.4 s	554

- Latency was reduced by ~11%, successfully eliminating the CPU processing bottleneck[2].
- The PDP worsened due to increased power consumption, as the Q6_K kernel fully utilizes all 64 PE units.

Conclusion

- We optimized Llama3 on the IMAX3 CGRA by offloading Q6_K operations with a new custom kernel, achieving a 11% reduction in latency.
- The IMAX3 architecture demonstrated superior energy efficiency (PDP) compared to standard GPU and CPU platforms.
- Future work will focus on tuning architectural parameters to further improve performance and efficiency.

References

1. T. Akabe, V. Trung Duong LE, and Y. Nakashima, "IMAX: A power-efficient multilevel pipelined cglra and applications," IEEE Access, vol. 13, pp. 31899–31911, 2025..

2. Y. Eto and Y. Nakashima, "Implementation and Performance Analysis of LLaMA on a CGLA," in International Conference on Intelligent Systems and Networks (ICISN), 2025..

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